UU-P-1144A
April 7, 1969
SUPERSEDING
Int. Fed. Spec. UU-P-001144 (GSA-FSS)
May 23, 1966 and
MIL-P-43351
July 28, 1965

FEDERAL SPECIFICATION

PAPER, WRAPPING, VEGETABLE PARCHMENT (FOR STERILIZATION)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. The document specifies the requirements for vegetable parchment wrapping paper suitable for use as a wrapping for instruments and dressings before, during, and after sterilization.

1.2 Sizes. Vegetable parchment paper shall be of one type of the following sizes or other sizes (in inches), as specified (see 6.2).

2-1/2 by 2-1/2
6 by 6
9 by 9
12 by 12
15 by 15
17 by 17
18 by 18
24 by 24
36 by 36

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifiations:

UU-P-268 - Paper, Kraft, Untreated, Wrapping.
NNN-P-35 - Paper, Blotting, Water, Laboratory.
PPP-B-636 - Box, Fiberboard.
PPP-T-76 - Tape, Pressure-Sensitive Adhesive Paper (For Carton Sealing).
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Goverment may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402)

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles and Seattle, Wash.)

(Federal Government activities may obtain copies of Federal Specifications, Standards and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:21Z
Check the source to verify that this is the current version before use.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

Technical Association of the Pulp and Paper Industry (TAPPI) Publications:

T402-os-49 - Conditioning Paper and Paperboard for Testing.
T403-ts-63 - Bursting Strength of Paper.
T404-ts-66 - Tensile Breaking Strength of Paper and Paperboard.
T410-os-61 - Basis Weight of Paper and Paperboard.
T411-os-44 - Thickness and Density of Paper.
T414-ts-65 - Internal Tearing Resistance of Paper.
T456-ts-49 - Wet Tensile Breaking Strength of Paper and Paperboard.
T435-ts-52 - Hydrogen Ion Concentrated (pH) of Paper Extracts.

(Copies of TAPPI Testing Methods may be obtained from the Technical Association of the Pulp and Paper Industry, 360 Lexington Avenue, New York, New York 10017.)

3. REQUIREMENTS

3.1 Material. The wrapping paper supplied in accordance with this specification shall be made from a 27 pound (24 by 36 - 500) plus or minus 5 percent vegetable parchment paper, treated with glycerine or other suitable non-volatile humectant.

3.2 Physical properties. The wrapping paper shall conform to the requirements of table I when tested as specified in 4.3.2.

TABLE I. Physical requirements

Property	Requirement	Tolerance
Basis weight (24 by 36 inches, 500 sheets) pounds, after humectant treatment	31	± 7.5 percent
Bursting strength, points, minimum		
Dry	17	
Wet	8.5	
Tearing resistance, grams, minimum		
Machine direction	20	
Cross machine direction	22.5	
Tensile breaking strength, pounds per inch width, minimum		
Dry		
Machine direction	16	
Cross machine direction	9	
Wet		
Machine direction	5.5	
Cross machine direction	3.6	
Thickness, inch	0.0023	± 10 percent

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:21Z
Check the source to verify that this is the current version before use.

3.3 Strength after autoclaving. The wrapping paper, after treatment in an autoclave shall not crack or disintegrate under normal handling conditions and shall have the following minimum wet bursting strength expressed in points, when tested as specified in 4.3.2:

Wet steam (under autoclaving)	8 points
Immersed in water (under autoclaving)	8 points

3.4 Chemical properties. The wrapping paper shall conform to the following requirements, when tested as specified in 4.3.2:

Acidity, pH value, minimum	4.5
Formaldehyde reaction	negative

3.5 Form and size. Vegetable parchment wrapping paper shall be furnished in sheet form in the size specified (see 6.2) with a dimensional tolerance of plus or minus 1/8-inch.

3.6 Workmanship. Vegetable parchment wrapping paper shall be clean and free from slime spots, holes, tears, folds or other imperfections. The sheets shall be clean cut without nicks or ragged edges and conform to the levels of quality established herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein, Except as otherwise specified, the supplier may use his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplier's services conform to prescribed requirements.

4.1.1 Sampling for inspection. Sampling for inspection shall be performed in accordance with the provision set forth in MIL-STD-105, except where otherwise indicated.

4.2 Material and components. The supplier is responsible for insuring that all materials and components used were inspected in accordance with the requirements of this specification and of referenced specifications.

4.2.1 Testing of the component. The supplier shall furnish a certificate of compliance stating that the basis weight of the vegetable parchment paper conforms to the requirements stated in 3.1 when tested by TAPPI Test T410-os-61.

4.3 Inspection of end item.

4.3.1 Examination of the end item. The end item shall be examined for the defects set forth in the applicable subparagraphs at the inspection levels and acceptable quality levels (AQL's) in 4.3.1.4. Random samples shall be drawn from each lot of the end item for examination of visual, dimensional, count, and preparation for delivery defects. The lot size, for the purposes of determining the sample size in accordance with MIL-STD-105 shall be expressed in terms of unit packages for examination in 4.3.1.1 and 4.3.1.2 and in units of shipping containers for examination under 4.3.1.3.

4.3.1.1 Examination for visual defects. The sample unit for this examination shall be one sheet. Both sides of the material shall be examined.

Examine	Defects
Appearance	Color not uniform. Not clean; any discoloration or mark affecting appearance or serviceability. Dirt spots, lint, dust, slivers, shives, holes, breaks, wrinkles, sharp creases, folds, cuts, or tears. Presence of foreign or undigested matter.
Workmanship	Edges not clean cut, nicked, or ragged.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:21Z
Check the source to verify that this is the current version before use.

4.3.1.2 Examination for dimensional defects. The sample unit for this examination shall be one sheet.

Examine	Defects
Length or width	Varies from dimension specified by more than ± 1/8-inch

4.3.1.3 Examination of preparation for delivery. An examination shall be made to determine that markings, materials, workmanship, count and contents comply with the requirements of section 5. The sample unit shall be one shipping container, fully packed, selected just prior to the closing operation. Closed shipping containers shall be examined for closure defects. The quality of the lot shall be unacceptable if the average count per intermediate package is less than specified when one intermediate package per shipping container is examined for count from the containers selected at inspection level S-2 in the preparation for delivery examination.

Examine	Defects
Marking (exterior and interior)	Incorrect; incomplete; illegible; omitted; of improper size, location, sequence, or method of application.
Materials	Any noncomforming component; component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of components such as; incomplete closure of container flaps, loose or inadequate sealing, strapping or stapling.
Contents	Number of intermediate packages per shipping container more or less than required.
Containers	Not as specified.

4.3.1.4 Inspection levels and acceptable quality levels (AQL's) for examination. The inspection levels for determining the sample size, and acceptable quality (AQL's) expressed in defects per 100 units shall be as follows:

Examination paragraph	Inspection level	AQL
4.3.1.1	II	1.5
4.3.1.2	S-4	4.0
4.3.1.3	S-2	2.5

4.3.2 Testing of the end item. The methods of testing specified in TAPPI, where applicable and as listed in table II shall be followed. The test sample unit shall consist of at least 1000 square inches of paper. Not more than one sample unit shall be drawn from any one package. The lot size shall be expressed in units of pounds of paper. 1/ The sample size shall be selected in accordance with the sampling plan which follows. Test results shall include all values upon which results are based and there shall be no failure to meet the sample unit, composite or lot average requirements. Composite samples shall be made up by combining small portions of approximately equal size from all sample units comprising the sample size.

1/ For the purpose of converting the lot size expressed in reams of a specified sheet size into pounds of paper, the following approximations may be used:

Sheet size	Weight per ream in pounds
2-1/2 by 2-1/2	0.22
6 by 6	1.3
9 by 9	2.9
12 by 12	5.2
15 by 15	8.1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:21Z
Check the source to verify that this is the current version before use.

Sheet size	Weight per ream in pounds
17 by 17	10.4
24 by 24	20.7
36 by 36	46.3

Sampling Plan

Lot size (pounds of paper)	Sample size
800 or less	2
801 to 22,000	3
22,001 or more	5

TABLE II. Instructions for testing of the end item

Characteristic	Specification reference: Requirement	Specification reference: TAPPI Test method	Requirements applicable to: Individ. Unit	Requirements applicable to: Lot Aver.	Number determination per sample unit	Results reported as: Pass or fail	Results reported as: Numerically to nearest
Atmospheric conditioning	4.3.2	T 402-os-49	All testing.				
Basis weight (after humectant treatment)	3.2.1	T 410-os-61	-	X	1 on 1000 sq. in. specimen.		0.1 pound
Thickness	3.2.1	T 411-os-44	-	X	Average of 10		0.0001 inch
Bursting strength:							
Dry	3.2.1	T 403-ts-63	-	X	10 (5 on each side)		point
Wet	3.2.1	4.4.1	-	X	10 (5 on each side)		point
Tearing resistance 1/							
Machine direction	3.2.1	T 414-ts-65	-	X	5		gram
Cross direction	3.2.1	T 414-ts-65	-	X	5		gram
Tensile strength 2/							
Dry							
Machine direction	3.2.1	T 404-os-61	-	X	10		0.1 pound
Cross direction	3.2.1	T 404-os-61	-	X	10		0.1 pound
Wet							
Machine direction	3.2.1	T 456-ts-49	-	X	10		0.1 pound
Cross direction	3.2.1	T-456-ts-49	-	X	10		0.1 pound
Strength after sterilization							
Water immersion	3.3	4.4.2	-	X	10 (5 on each side)		point
Wet steam	3.3	4.4.2	-	X	10 (5 on each side)		point
Acidity (pH)	3.4	T 435-ts-52 3/	-	-	2 on composite		0.1 pH
Formaldehyde reaction	3.4	4.4.3	X	-	1	X	

1/ For samples of the 2 1/2 by 2 1/2-inch paper, the specimen length for the tearing resistance test shall be 2 1/2-inches.

2/ For samples of the 2 1/2 by 2 1/2-inch paper, the jaws of the tensile tester shall be set 1-inch apart.

3/ The hot extraction method of T 435-ts-52 shall be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:21Z
Check the source to verify that this is the current version before use.

4.4 Tests.

4.4.1 Wet bursting strength. Heat the test specimens selected for this test at 105° ± 2°C for 15 minutes and immediately soak in distilled water for one hour at a temperature of 23° ± 2°C. Remove the specimens, drain the excess water and blot lightly between two sheets of blotting paper conforming to NNN-P-35. Test the specimens immediately thereafter for bursting strength in accordance with T 403-ts-63.

4.4.2 Strength after autoclaving.

4.4.2.1 Apparatus. An autoclave capable of being operated at a pressure of 15 to 17 pounds per square inch steam pressure and at an exhaust temperature of 121°C (249.8°F).

4.4.2.2 Specimens. Cut ten (10) specimens, 2.5-inches by 2.5-inches, for each of the two phases of this test, taking care to mark the specimens in order to identify the felt and wire sides. The specimens shall be cut from sheets that have been conditioned as specified.

4.4.2.3 Procedure, water immersion. Place the specimens in a suitably sized screw cap bottle, not more that five in one container, and fill the bottle with distilled water. Tighten the screw cap and place the bottle in the autoclave. Adjust the autoclave so as to maintain a steam pressure of 15 to 17 pounds pressure and an exhaust temperature of 121°C (249.8°F). Allow the containers to remain in the autoclave for a period of 90 ± 5 minutes. At the end of this time, release the autoclave pressure and allow the specimens and containers to reach room temperature. Remove the specimens, drain the excess water and blot lightly with blotting paper conforming to NNN-P-35. Test the specimens immediately thereafter for bursting strength in accordance with T 403-ts-63.

4.4.2.4 Procedure, wet steam. Place the specimens, prepared as specified in 4.4.2.2 on a wire screen in the autoclave. Adjust the autoclave to the temperature and pressure specified in 4.4.2.3 and allow the specimens to remain in the autoclave for 90 ± 5 minutes. At the end of this time, release the autoclave pressure and remove the samples. Remove any water drops by blotting lightly as in 4.4.2.3 and immediately thereafter, determine the bursting strength in accordance with T 403-ts-63.

4.4.3 Formaldehyde reaction. The formaldehyde reaction is performed in two steps. In the first step, the resins in the wet-strength (treated) paper are hydrolyzed by an acid solution to liberate formaldehyde. In the final step, another chemical is added. This chemical develops a characteristic color if the resins are present.

4.4.3.1 Reagents required. Two solutions referred to as solution I and solution II, are required. They are prepared as follows:

Solution I - Dissolve 1.34 grams of phenylhydrazene hydrochloride crystal (or 1.0 gram of phenylhydrazene) in 41.7 grams of concentrated (96 percent) sulfuric acid, which has been diluted to about 80 grams with water. This solution is then diluted with additional water to 100 grams total.

Solution II - Dissolve 10 grams of anhydrous ferric chloride (or 16 grams of $FeCl_3 6H_2O$) in water sufficient to make 100 grams solution.

4.4.3.2 Test procedure. Using an uncontaminated glass rod or dropper, spread a drop of solution I on the paper to be tested over an area of approximately 1-inch in diameter. After 30 seconds, remove any excess liquid with a blotter and add a drop of solution II. If either melamine formaldehyde or urea-formaldehyde resins are present, a pink, red, or reddish brown color will develop.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. Five hundred (500) sheets of paper of like description shall be wrapped in kraft paper conforming to UU-P-268, grade B and all folds and laps closed with tape conforming to PPP-T-76.

5.1.2 Level B. Packaging shall be as specified in paragraph 5.1.1.

5.1.3 Level C. Wrapping paper, of one size only, shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. The unit packages of sheets in quantities as specifed in Table III shall be packed in a close-fitting box conforming to PPP-B-636, grade V2s. The box shall be closed, waterproofed sealed and reinforced with flat strapping or tape banding in accordance with appendix to the box specification.

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:21Z
Check the source to verify that this is the current version before use.